

Data Analysis & Probability

ANSWER KEY



Mean, median, mode and range

- 1 Find the mean of 6, 9, 12, 15, 18.

$$\text{Mean} = \frac{6 + 9 + 12 + 15 + 18}{5} = \frac{60}{5} = 12.$$

- 2 Ages of 7 students: 13, 15, 13, 16, 14, 13, 17. Find the mean, median, and mode.

Ordered: 13, 13, 13, 14, 15, 16, 17. Mean = $\frac{101}{7} \approx 14.43$. Median (4th value) = 14. Mode = 13 (appears three times).

- 3 Find the range and mode of the data set 8, 3, 8, 15, 8, 22, 3.

Range = $22 - 3 = 19$. Mode = 8 (appears three times).

- 4 A data set has mean 18 and 6 values. Find the sum of the values.

$$\text{Sum} = 18 \times 6 = 108.$$

- 5 The data set 10, 12, 11, 13, 50 contains an outlier. Find the mean before and after removing 50.

Before: $\frac{10 + 12 + 11 + 13 + 50}{5} = \frac{96}{5} = 19.2$. After removing 50: $\frac{46}{4} = 11.5$.

- 6 A data set of 5 values has mean 20. Four of the values are 15, 18, 22, 25. Find the fifth value.

Total sum = $20 \times 5 = 100$. Known sum = $15 + 18 + 22 + 25 = 80$. Fifth value = $100 - 80 = 20$.

Reading tables

- 7 A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?

$$\frac{12}{50} = \frac{6}{25}.$$

- 8 Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?

$$15 + 13 = 28 \text{ people. } \frac{28}{50} \times 100\% = 56\%.$$

- 9 A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.

Mode = 8 (8 students, the highest frequency). Total students = $3 + 5 + 8 + 4 + 2 = 22$.

- 10 Using the same quiz data, find the mean score to two decimal places.

$$\text{Sum} = 5(3) + 7(5) + 8(8) + 9(4) + 10(2) = 15 + 35 + 64 + 36 + 20 = 170. \text{ Mean} = \frac{170}{22} \approx 7.73.$$

Reading bar charts

- 11 The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.

Total = $45 + 30 + 20 + 15 = 110$ books. The modal genre is Fiction, with 45 books.

- 12 Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?

$$\frac{30}{110} \times 100\% \approx 27\%.$$

- 13 The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.

$$\text{Mean} = \frac{60 + 45 + 80 + 55}{4} = \frac{240}{4} = 60 \text{ mm}.$$

- 14 Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.

$$\text{Decrease} = 80 - 55 = 25 \text{ mm}. \frac{25}{80} \times 100\% = 31.25\%.$$

Basic probability

- 15 A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.

$$P(\text{red}) = \frac{4}{10} = \frac{2}{5}.$$

- 16 A fair six-sided die is rolled. Find $P(\text{a number greater than 4})$.

$$\text{Favorable outcomes } \{5, 6\}: P = \frac{2}{6} = \frac{1}{3}.$$

- 17 A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.

$$P = \frac{13}{52} = \frac{1}{4}.$$

- 18 A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.

$$\text{Even outcomes } \{2, 4, 6, 8\}: P = \frac{4}{8} = \frac{1}{2}.$$

Combined and complementary events

- 19 A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.

$$P(\text{red}) = \frac{3}{10}, \text{ so } P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}.$$

- 20 Two fair coins are tossed. Find $P(\text{both land heads})$.

$$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

- 21 A die is rolled twice. Find $P(\text{both rolls show a 6})$.

$$P = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}.$$

- 22 A bag holds 4 white and 6 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both white})$.

$$P = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}.$$

Weighted averages

- 23 A class of 9 students has an average score of 80. A 10th student joins with a score of 100. Find the new average for all 10 students.

$$\text{Old sum} = 80 \times 9 = 720. \text{ New sum} = 720 + 100 = 820. \text{ New average} = \frac{820}{10} = 82.$$

- 24 Class A has 20 students with mean score 75; Class B has 30 students with mean score 85. Find the overall mean of all 50 students.

$$\text{Total sum} = 20(75) + 30(85) = 1,500 + 2,550 = 4,050. \text{ Overall mean} = \frac{4,050}{50} = 81.$$

- 25 A student's final grade is 40% homework average (score 90) and 60% exam average (score 70). Find the weighted final grade.

$$0.4 \times 90 + 0.6 \times 70 = 36 + 42 = 78.$$

- 26 A survey combines two groups: 15 people with average age 24, and 25 people with average age 32. Find the overall average age of all 40 people.

$$\text{Total sum} = 15(24) + 25(32) = 360 + 800 = 1,160. \text{ Overall mean} = \frac{1,160}{40} = 29.$$

Comparing data sets

- 27 Data set A: 10, 12, 14, 16, 18. Data set B: 8, 14, 14, 14, 20. Both have the same mean. Which has the greater range?

Both have mean 14. Range of A = $18 - 10 = 8$. Range of B = $20 - 8 = 12$. Data set B has the greater range (more spread out).

- 28** Team X scored 22, 25, 28, 25 points in 4 games. Team Y scored 20, 20, 35, 25. Both teams have the same mean score. Which team's scores are more consistent (smaller range)?

Both have mean 25. Range of X = $28 - 22 = 6$. Range of Y = $35 - 20 = 15$. Team X is more consistent.

- 29** Class 1 has 18 students with mean score 68; Class 2 has 22 students with mean score 74. Find the combined mean of all 40 students, to one decimal place.

Total sum = $18(68) + 22(74) = 1,224 + 1,628 = 2,852$. Combined mean = $\frac{2,852}{40} = 71.3$.

Word problems

- 30** A teacher records the number of siblings for 10 students: 1, 2, 0, 1, 3, 2, 1, 0, 2, 1. Find the mean, median, and mode.

Sum = 13, mean = $\frac{13}{10} = 1.3$. Ordered: 0, 0, 1, 1, 1, 1, 2, 2, 2, 3; median = $\frac{1 + 1}{2} = 1$. Mode = 1 (appears four times).

- 31** A company's weekly sales (in thousands of riyals) over 5 weeks: 40, 55, 35, 60, 50. Find the mean weekly sales, and state how many weeks had sales above the mean.

Mean = $\frac{240}{5} = 48$ thousand riyals. Weeks above the mean: 55, 60, 50 — that is 3 weeks.

- 32** A jar contains 5 red, 3 blue and 2 yellow candies. A candy is picked and eaten, then another is picked. Find $P(\text{first red, then blue})$.

$$P = \frac{5}{10} \times \frac{3}{9} = \frac{15}{90} = \frac{1}{6}.$$

- 33** In a survey of 200 students, 120 prefer math and the rest prefer science. Find the probability, as a percent, that a randomly chosen student prefers science.

Science = $200 - 120 = 80$ students. $P = \frac{80}{200} = 0.4 = 40\%$.

MCQ — Mean, median, mode and range

34 Find the mean of 8, 11, 14, 17, 20.

- a. 12
- b. 17
- c. 70
- d. 14 ✓

Sum = 70; mean = $70 \div 5 = 14$.

35 Ages of 6 students: 12, 14, 12, 15, 13, 12. Find the mean, median, and mode.

- a. Mean = 13, median = 12, mode = 12.5
- b. Mean = 13, median = 12.5, mode = 12 ✓
- c. Mean = 12.5, median = 13, mode = 12
- d. Mean = 13, median = 12.5, mode = 13

Sum = 78, mean = 13. Sorted: 12, 12, 12, 13, 14, 15; median = $\frac{12 + 13}{2} = 12.5$; mode = 12 (appears 3 times).

36 Find the range and mode of the data set 5, 9, 5, 18, 5, 27, 9.

- a. Range = 13, mode = 5
- b. Range = 27, mode = 5
- c. Range = 22, mode = 5 ✓
- d. Range = 22, mode = 9

Range = $27 - 5 = 22$; mode = 5 (appears 3 times).

37 A data set has mean 24 and 8 values. Find the sum of the values.

- a. 32
- b. 3
- c. 192 ✓
- d. 96

$$\text{Sum} = \text{mean} \times \text{count} = 24 \times 8 = 192.$$

38 The data set 14, 16, 15, 17, 60 contains an outlier. Find the mean before and after removing 60.

- a. Before = 122, after = 15.5
- b. Before = 24.4, after = 15.5 ✓
- c. Before = 15.5, after = 24.4
- d. Before = 24.4, after = 62

Before: sum = 122, mean = 24.4. After removing 60: sum = 62 over 4 values, mean = 15.5.

39 A data set of 6 values has mean 25. Five of the values are 20, 22, 28, 30, 32. Find the sixth value.

- a. 18 ✓
- b. 28
- c. 8
- d. 150

$$\text{Total needed} = 6 \times 25 = 150; \text{sum of five} = 132; \text{sixth} = 150 - 132 = 18.$$

40 The bar chart shows quiz scores. Find the total number of students, and the modal score.

- a. 20 students; mode = 8 ✓
- b. 20 students; mode = 6
- c. 20 students; mode = 9
- d. 18 students; mode = 8

Total = 2 + 4 + 6 + 5 + 3 = 20; the score 8 has the highest frequency (6 students).

MCQ — Reading tables

- 41 A survey of 60 people's favorite drink: Tea 18, Coffee 22, Juice 12, Water 8. What fraction of people chose coffee?

- a. $\frac{18}{60}$
- b. $\frac{22}{60}$
- c. $\frac{11}{30}$ ✓
- d. $\frac{11}{60}$

$\frac{22}{60}$ simplifies (divide by 2) to $\frac{11}{30}$.

- 42 Using the same survey (Tea 18, Coffee 22, Juice 12, Water 8, out of 60), what percent chose either tea or water, to the nearest percent?

- a. 43% ✓
- b. 30%
- c. 50%
- d. 26%

$$(18 + 8) \div 60 = 26 \div 60 \approx 43\%.$$

- 43 A table shows test scores: score 60 (4 students), 70 (7), 80 (9), 90 (6), 100 (4). Find the total number of students.

- a. 34
- b. 30 ✓
- c. 26
- d. 25

$$4 + 7 + 9 + 6 + 4 = 30.$$

44 Using the same test data, find the mean score to two decimal places.

a. 79.67 ✓

b. 80.00

c. 76.33

d. 79.00

Sum = $240 + 490 + 720 + 540 + 400 = 2,390$; mean = $2,390 \div 30 \approx 79.67$.

45 A table shows the number of pets owned by 25 families: 0 pets (6 families), 1 pet (10), 2 pets (7), 3 pets (2). Find the median number of pets.

a. 0

b. 1.2

c. 2

d. 1 ✓

With 25 values sorted, the median is the 13th value, which falls in the '1 pet' group.

46 Using the same pet data, find the mean number of pets per family, to two decimal places.

a. 1.00

b. 1.20 ✓

c. 1.50

d. 30

Sum = $0(6) + 1(10) + 2(7) + 3(2) = 30$; mean = $30 \div 25 = 1.20$.

MCQ — Reading bar charts

47 The bar chart shows books sold by genre in one week. Find the total books sold, and the modal genre.

- a. 120; Romance
- b. 100; Fiction
- c. 120; Mystery
- d. 120; Fiction ✓

Total = $45 + 30 + 25 + 20 = 120$; Fiction had the most sales (45).

48 Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?

- a. 25% ✓
- b. 30%
- c. 20%
- d. 38%

$30 \div 120 = 25\%$.

49 The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.

- a. 225
- b. 65
- c. 55
- d. 56.25 ✓

$$\text{Mean} = \frac{50 + 70 + 40 + 65}{4} = \frac{225}{4} = 56.25 \text{ mm.}$$

- 50 Using the same rainfall data (Jan 50, Feb 70, Mar 40, Apr 65 mm), find the percent decrease from Feb to Mar, to the nearest percent.

- a. 75%
- b. 43% ✓
- c. 30%
- d. 40%

Decrease = $70 - 40 = 30$; $30 \div 70 \approx 43\%$.

- 51 The bar chart shows quarterly sales for four stores, in thousands of riyals. Find the range, and which store had the highest sales.

- a. 55; Store B
- b. 55; Store D ✓
- c. 140; Store D
- d. 35; Store D

Range = $140 - 85 = 55$; Store D had the highest sales (140).

- 52 The bar chart shows hours studied each weekday. Find the total hours, and the mean hours per day.

- a. 12; 2
- b. 12; 2.4 ✓
- c. 12; 3
- d. 10; 2

Total = $2 + 3 + 1 + 4 + 2 = 12$; mean = $12 \div 5 = 2.4$.

- 53 The bar chart shows the number of defective items found for four products in a quality check. Find the total defects, and the range.

a. 40; 10 ✓

b. 40; 7

c. 35; 10

d. 40; 15

Total = $12 + 8 + 15 + 5 = 40$; range = $15 - 5 = 10$.

MCQ — Basic probability

- 54 A bag holds 5 red and 7 blue marbles. Find $P(\text{red})$.

a. $\frac{7}{12}$

b. $\frac{5}{12}$ ✓

c. $\frac{5}{7}$

d. $\frac{1}{12}$

5 red out of 12 total.

- 55 A fair six-sided die is rolled. Find $P(\text{a number less than 3})$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{2}{3}$

d. $\frac{1}{3}$ ✓

Favorable outcomes $\{1, 2\}$: $\frac{2}{6} = \frac{1}{3}$.

56 A card is drawn from a standard deck of 52 cards. Find $P(\text{a king})$.

- a. $\frac{1}{4}$
- b. $\frac{1}{52}$
- c. $\frac{1}{13}$ ✓
- d. $\frac{4}{13}$

4 kings out of 52: $\frac{4}{52} = \frac{1}{13}$.

57 A spinner has 10 equal sections numbered 1 to 10. Find $P(\text{a multiple of 3})$.

- a. $\frac{1}{10}$
- b. $\frac{1}{3}$
- c. $\frac{4}{10}$
- d. $\frac{3}{10}$ ✓

Multiples of 3 from 1–10: $\{3, 6, 9\}$, so $\frac{3}{10}$.

58 A bag holds 6 yellow and 9 green balls. Find $P(\text{yellow})$.

- a. $\frac{1}{5}$
- b. $\frac{6}{9}$
- c. $\frac{3}{5}$
- d. $\frac{2}{5}$ ✓

6 out of 15 simplifies to $\frac{2}{5}$.

59 A fair six-sided die is rolled. Find $P(\text{an odd number})$.

- a. $\frac{2}{3}$
- b. $\frac{1}{6}$
- c. $\frac{1}{2}$ ✓
- d. $\frac{1}{3}$

Odd outcomes $\{1, 3, 5\}$: $\frac{3}{6} = \frac{1}{2}$.

MCQ — Combined and complementary events

60 A bag holds 4 red, 6 blue and 5 green marbles. Find $P(\text{not blue})$.

- a. $\frac{3}{5}$ ✓
- b. $\frac{9}{6}$
- c. $\frac{2}{5}$
- d. $\frac{6}{15}$

Not blue = 4 + 5 = 9 out of 15: $\frac{9}{15} = \frac{3}{5}$.

61 Two fair coins are tossed. Find $P(\text{at least one tail})$.

- a. $\frac{3}{4}$ ✓
- b. $\frac{1}{3}$
- c. $\frac{1}{4}$
- d. $\frac{1}{2}$

$P(\text{no tails}) = \frac{1}{4}$, so $P(\text{at least one}) = 1 - \frac{1}{4} = \frac{3}{4}$.

62 A die is rolled twice. Find $P(\text{both rolls show an even number})$.

- a. $\frac{1}{6}$
- b. $\frac{1}{3}$
- c. $\frac{1}{4}$ ✓
- d. $\frac{1}{2}$

Each roll: $P(\text{even}) = \frac{1}{2}$; both: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

- 63 A bag holds 5 white and 7 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both black})$.

a. $\frac{7}{22}$ ✓

b. $\frac{6}{11}$

c. $\frac{49}{144}$

d. $\frac{7}{12}$

$$\frac{7}{12} \times \frac{6}{11} = \frac{42}{132} = \frac{7}{22}.$$

- 64 A card is drawn, replaced, then a second card is drawn from a standard deck. Find $P(\text{both hearts})$.

a. $\frac{1}{8}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$

d. $\frac{1}{16}$ ✓

$$\left(\frac{13}{52}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$$

- 65 A bag holds 3 red and 4 blue marbles. Two are drawn without replacement. Find $P(\text{both red})$.

a. $\frac{3}{7}$

b. $\frac{9}{49}$

c. $\frac{2}{7}$

d. $\frac{1}{7}$ ✓

$$\frac{3}{7} \times \frac{2}{6} = \frac{6}{42} = \frac{1}{7}.$$

MCQ — Weighted averages

- 66 A class of 12 students has an average score of 78. A 13th student joins with a score of 91. Find the new average.
- a. 84.50
 - b. 78.00
 - c. 91.00
 - d. 79.00 ✓

New total = $12(78) + 91 = 1,027$; new average = $1,027 \div 13 = 79.00$.

- 67 Class A has 25 students with mean score 70; Class B has 15 students with mean score 90. Find the combined mean.
- a. 80
 - b. 75
 - c. 160
 - d. 77.5 ✓

$$\frac{25(70) + 15(90)}{40} = \frac{3,100}{40} = 77.5.$$

- 68 A student's final grade is 30% homework average (score 85) and 70% exam average (score 75). Find the final grade.
- a. 82
 - b. 80
 - c. 78 ✓
 - d. 75.5

$$0.3(85) + 0.7(75) = 25.5 + 52.5 = 78.$$

- 69 A survey combines two groups: 20 people with average age 30, and 30 people with average age 40. Find the combined average age.

- a. 70
- b. 35
- c. 36 ✓
- d. 34

$$\frac{20(30) + 30(40)}{50} = \frac{1,800}{50} = 36.$$

- 70 A class of 8 students has an average score of 65. A 9th student joins, changing the new average to 68. Find the 9th student's score.

- a. 71
- b. 65
- c. 88
- d. 92 ✓

$$\text{New total} = 9(68) = 612; \text{old total} = 8(65) = 520; \text{9th score} = 612 - 520 = 92.$$

- 71 Two classes combine: Class A (10 students, mean 60) and Class B (n students, mean 90), giving a combined mean of 78 for all students. Find n .

- a. 12
- b. 15 ✓
- c. 10
- d. 20

$$\frac{600 + 90n}{10 + n} = 78 \Rightarrow 600 + 90n = 780 + 78n \Rightarrow 12n = 180 \Rightarrow n = 15.$$

MCQ — Comparing data sets

- 72 Data set A: 12, 14, 16, 18, 20. Data set B: 8, 15, 17, 19, 21. Both have the same mean. Which has the greater range?
- a. Cannot be determined
 - b. They are equal
 - c. Data set A
 - d. Data set B ✓

Both have mean 16. Range A = 8; range B = 13. B has the greater range.

- 73 Team X scored 18, 22, 25, 23 points in 4 games. Team Y scored 15, 15, 32, 26. Both teams have the same mean. Which team is more consistent (smaller range)?
- a. Team X ✓
 - b. Team Y
 - c. Cannot be determined
 - d. They are equally consistent

Both have mean 22. Range X = 7; range Y = 17. Team X is more consistent.

- 74 Class 1 has 16 students with mean score 70; Class 2 has 24 students with mean score 80. Find the combined mean of both classes.
- a. 150
 - b. 75
 - c. 76 ✓
 - d. 77

$$\frac{16(70) + 24(80)}{40} = \frac{3,040}{40} = 76.$$

- 75 Data set A: 5, 10, 15, 20, 25. Data set B: 13, 14, 15, 16, 17. Both have mean 15. Which has the smaller range, and what is it?
- a. Data set B; 20
 - b. Data set B; 4 ✓
 - c. Data set A; 4
 - d. Data set A; 20

Range A = 20; range B = 4. Data set B is smaller.

- 76 Two data sets both have median 20. Set P: 10, 15, 20, 25, 30. Set Q: 18, 19, 20, 21, 22. Which set is more spread out?
- a. Set P ✓
 - b. Cannot be determined
 - c. Set Q
 - d. They are equally spread out

Range P = 20; range Q = 4. Set P is more spread out.

MCQ — Word problems

- 77 A teacher records the number of siblings for 12 students: 2, 1, 0, 2, 3, 1, 2, 0, 1, 2, 3, 1. Find the mean, to two decimal places.
- a. 1.33
 - b. 1.60
 - c. 1.50 ✓
 - d. 2.00

Sum = 18; mean = $18 \div 12 = 1.50$.

- 78 A company's weekly sales (in thousands of riyals) over 6 weeks: 45, 60, 38, 72, 55, 50. Find the mean weekly sales.
- a. 55.00
 - b. 320
 - c. 50.00
 - d. 53.33 ✓

Sum = 320; mean = $320 \div 6 \approx 53.33$.

- 79 A jar contains 6 red, 4 blue and 2 yellow candies. A candy is picked and eaten, then another is picked. Find $P(\text{both red})$.

- a. $\frac{6}{12}$
- b. $\frac{5}{22}$ ✓
- c. $\frac{1}{4}$
- d. $\frac{3}{11}$

$$\frac{6}{12} \times \frac{5}{11} = \frac{30}{132} = \frac{5}{22}.$$

- 80 In a survey of 300 students, 180 prefer math and the rest prefer science. Find the probability, as a percent, that a randomly chosen student prefers science.
- a. 120%
 - b. 40% ✓
 - c. 18%
 - d. 60%

Science = 120 out of 300: 40%.

- 81 The bar chart shows a company's quarterly sales, in thousands of riyals. Find the total annual sales, and the mean quarterly sales.
- a. 1,560; 450
 - b. 1,560; 390 ✓
 - c. 1,560; 400
 - d. 1,200; 390

Total = $320 + 410 + 380 + 450 = 1,560$; mean = $1,560 \div 4 = 390$.

- 82 A survey of 150 shoppers found 90 prefer online shopping and the rest prefer in-store. Find the probability, as a fraction in simplest form, that a randomly chosen shopper prefers in-store.
- a. $\frac{2}{5}$ ✓
 - b. $\frac{90}{150}$
 - c. $\frac{3}{5}$
 - d. $\frac{3}{50}$

In-store = 60 out of 150, which simplifies to $\frac{2}{5}$.

- 83 A basketball player makes 18 out of 25 free throws attempted. Find the probability, as a decimal, that the next free throw is missed (based on this rate).
- a. 0.28 ✓
 - b. 0.18
 - c. 0.25
 - d. 0.72

Missed rate = $\frac{25 - 18}{25} = \frac{7}{25} = 0.28$.

Multiple choice — easy

84 Find the mean of 29, 23, 43, 25, 50.

- a. 32
- b. 34 ✓
- c. 39
- d. 36

$$\text{Mean} = \frac{29 + 23 + 43 + 25 + 50}{5} = \frac{170}{5} = 34.$$

85 Find the mean of 21, 31, 24, 38, 51.

- a. 38
- b. 35
- c. 33 ✓
- d. 31

$$\text{Mean} = \frac{21 + 31 + 24 + 38 + 51}{5} = \frac{165}{5} = 33.$$

86 Find the mean of 1, 12, 36, 40, 36.

- a. 27
- b. 25 ✓
- c. 30
- d. 23

$$\text{Mean} = \frac{1 + 12 + 36 + 40 + 36}{5} = \frac{125}{5} = 25.$$

87 Find the mean of 24, 23, 48, 18, 12.

- a. 30
- b. 25 ✓
- c. 23
- d. 27

$$\text{Mean} = \frac{24 + 23 + 48 + 18 + 12}{5} = \frac{125}{5} = 25.$$

88 Find the mean of 12, 35, 40, 42, 46.

- a. 40
- b. 33
- c. 37
- d. 35 ✓

$$\text{Mean} = \frac{12 + 35 + 40 + 42 + 46}{5} = \frac{175}{5} = 35.$$

89 Find the mean of 8, 49, 10, 47, 41.

- a. 33
- b. 36
- c. 31 ✓
- d. 29

$$\text{Mean} = \frac{8 + 49 + 10 + 47 + 41}{5} = \frac{155}{5} = 31.$$

90 Find the mean of 12, 4, 39, 43, 47.

- a. 27
- b. 31
- c. 29 ✓
- d. 34

$$\text{Mean} = \frac{12 + 4 + 39 + 43 + 47}{5} = \frac{145}{5} = 29.$$

91 Find the mean of 1, 41, 48, 60, 10.

- a. 37
- b. 30
- c. 34
- d. 32 ✓

$$\text{Mean} = \frac{1 + 41 + 48 + 60 + 10}{5} = \frac{160}{5} = 32.$$

92 Find the mean of 50, 50, 43, 21, 21.

- a. 39
- b. 42
- c. 35
- d. 37 ✓

$$\text{Mean} = \frac{50 + 50 + 43 + 21 + 21}{5} = \frac{185}{5} = 37.$$

93 Find the mean of 59, 18, 42, 1, 55.

- a. 40
- b. 37
- c. 35 ✓
- d. 33

$$\text{Mean} = \frac{59 + 18 + 42 + 1 + 55}{5} = \frac{175}{5} = 35.$$

94 Find the mean of 13, 41, 14, 6, 1.

- a. 20
- b. 15 ✓
- c. 17
- d. 13

$$\text{Mean} = \frac{13 + 41 + 14 + 6 + 1}{5} = \frac{75}{5} = 15.$$

95 Find the mean of 58, 36, 27, 34, 20.

a. 33

b. 35 ✓

c. 40

d. 37

$$\text{Mean} = \frac{58 + 36 + 27 + 34 + 20}{5} = \frac{175}{5} = 35.$$

96 Find the mean of 36, 5, 59, 57, 23.

a. 36 ✓

b. 34

c. 41

d. 38

$$\text{Mean} = \frac{36 + 5 + 59 + 57 + 23}{5} = \frac{180}{5} = 36.$$

97 Find the mean of 55, 7, 53, 41, 49.

a. 39

b. 46

c. 41 ✓

d. 43

$$\text{Mean} = \frac{55 + 7 + 53 + 41 + 49}{5} = \frac{205}{5} = 41.$$

98 Find the mean of 11, 48, 10, 37, 34.

a. 30

b. 28 ✓

c. 26

d. 33

$$\text{Mean} = \frac{11 + 48 + 10 + 37 + 34}{5} = \frac{140}{5} = 28.$$

Multiple choice — medium

99 A bag holds 3 yellow and 3 green marbles. Find $P(\text{yellow})$.

a. $\frac{2}{3}$

b. $\frac{5}{8}$

c. $\frac{1}{2}$ ✓

d. $\frac{5}{6}$

$$P(\text{yellow}) = \frac{3}{6} = \frac{1}{2}.$$

100 A bag holds 7 red and 2 blue marbles. Find $P(\text{red})$.

a. $\frac{8}{9}$

b. $\frac{7}{9}$ ✓

c. $\frac{2}{9}$

d. $\frac{9}{11}$

$$P(\text{red}) = \frac{7}{9}.$$

101 A bag holds 3 yellow and 9 green marbles. Find $P(\text{yellow})$.

a. $\frac{1}{4}$ ✓

b. $\frac{1}{3}$

c. $\frac{5}{14}$

d. $\frac{3}{4}$

$$P(\text{yellow}) = \frac{3}{12} = \frac{1}{4}.$$

102 A bag holds 8 red and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{8}{17}$ ✓

b. $\frac{10}{17}$

c. $\frac{9}{17}$

d. $\frac{10}{19}$

$$P(\text{red}) = \frac{8}{17}.$$

103 A bag holds 2 yellow and 11 green marbles. Find $P(\text{yellow})$.

a. $\frac{3}{13}$

b. $\frac{11}{13}$

c. $\frac{2}{13}$ ✓

d. $\frac{4}{15}$

$$P(\text{yellow}) = \frac{2}{13}.$$

104 A bag holds 6 red and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{3}{5}$

b. $\frac{8}{17}$

c. $\frac{2}{5}$ ✓

d. $\frac{7}{15}$

$$P(\text{red}) = \frac{6}{15} = \frac{2}{5}.$$

105 A bag holds 3 black and 11 white marbles. Find $P(\text{black})$.

a. $\frac{5}{16}$

b. $\frac{2}{7}$

c. $\frac{3}{14}$ ✓

d. $\frac{11}{14}$

$$P(\text{black}) = \frac{3}{14}.$$

106 A bag holds 9 black and 5 white marbles. Find $P(\text{black})$.

a. $\frac{5}{14}$

b. $\frac{11}{16}$

c. $\frac{5}{7}$

d. $\frac{9}{14}$ ✓

$$P(\text{black}) = \frac{9}{14}.$$

107 A bag holds 11 red and 5 blue marbles. Find $P(\text{red})$.

a. $\frac{5}{16}$

b. $\frac{3}{4}$

c. $\frac{13}{18}$

d. $\frac{11}{16}$ ✓

$$P(\text{red}) = \frac{11}{16}.$$

108 A bag holds 5 yellow and 12 green marbles. Find $P(\text{yellow})$.

a. $\frac{6}{17}$

b. $\frac{5}{17}$ ✓

c. $\frac{12}{17}$

d. $\frac{7}{19}$

$$P(\text{yellow}) = \frac{5}{17}.$$

109 A bag holds 9 black and 2 white marbles. Find $P(\text{black})$.

a. $\frac{11}{13}$

b. $\frac{2}{11}$

c. $\frac{9}{11}$ ✓

d. $\frac{10}{11}$

$$P(\text{black}) = \frac{9}{11}.$$

110 A bag holds 10 yellow and 10 green marbles. Find $P(\text{yellow})$.

a. $\frac{6}{11}$

b. $\frac{1}{2}$ ✓

c. $\frac{11}{20}$

d. $\frac{3}{5}$

$$P(\text{yellow}) = \frac{10}{20} = \frac{1}{2}.$$

111 A bag holds 8 black and 5 white marbles. Find $P(\text{black})$.

a. $\frac{2}{3}$

b. $\frac{9}{13}$

c. $\frac{5}{13}$

d. $\frac{8}{13}$ ✓

$$P(\text{black}) = \frac{8}{13}.$$

112 A bag holds 11 red and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{13}{22}$

b. $\frac{9}{20}$

c. $\frac{11}{20}$ ✓

d. $\frac{3}{5}$

$$P(\text{red}) = \frac{11}{20}.$$

113 A bag holds 11 red and 6 blue marbles. Find $P(\text{red})$.

a. $\frac{12}{17}$

b. $\frac{6}{17}$

c. $\frac{13}{19}$

d. $\frac{11}{17}$ ✓

$$P(\text{red}) = \frac{11}{17}.$$

Multiple choice — hard

114 Class A has 30 students with mean score 61; Class B has 32 students with mean score 92. Find the overall mean of all 62 students.

a. 76

b. 77 ✓

c. 75

d. 79

$$\text{Total sum} = 30(61) + 32(92) = 1830 + 2944 = 4774. \text{ Overall mean} = \frac{4774}{62} = 77.$$

115 Class A has 21 students with mean score 93; Class B has 27 students with mean score 61. Find the overall mean of all 48 students.

- a. 77
- b. 78
- c. 73
- d. 75 ✓

$$\text{Total sum} = 21(93) + 27(61) = 1953 + 1647 = 3600. \text{ Overall mean} = \frac{3600}{48} = 75.$$

116 Class A has 23 students with mean score 83; Class B has 23 students with mean score 81. Find the overall mean of all 46 students.

- a. 82 ✓
- b. 84
- c. 85
- d. 80

$$\text{Total sum} = 23(83) + 23(81) = 1909 + 1863 = 3772. \text{ Overall mean} = \frac{3772}{46} = 82.$$

117 Class A has 26 students with mean score 65; Class B has 22 students with mean score 89. Find the overall mean of all 48 students.

- a. 78
- b. 76 ✓
- c. 77
- d. 74

$$\text{Total sum} = 26(65) + 22(89) = 1690 + 1958 = 3648. \text{ Overall mean} = \frac{3648}{48} = 76.$$

118 Class A has 35 students with mean score 74; Class B has 35 students with mean score 94. Find the overall mean of all 70 students.

- a. 87
- b. 82
- c. 86
- d. 84 ✓

$$\text{Total sum} = 35(74) + 35(94) = 2590 + 3290 = 5880. \text{ Overall mean} = \frac{5880}{70} = 84.$$

119 Event X has probability $\frac{1}{5}$ and independent event Y has probability $\frac{4}{6}$. Find $P(X \text{ and } Y)$.

- a. $\frac{13}{15}$
- b. $\frac{7}{30}$
- c. $\frac{2}{15}$ ✓
- d. $\frac{1}{30}$

$$P(X \text{ and } Y) = \frac{1}{5} \times \frac{4}{6} = \frac{4}{30} = \frac{2}{15}.$$

120 Event X has probability $\frac{1}{8}$ and independent event Y has probability $\frac{4}{8}$. Find $P(X \text{ and } Y)$.

- a. $\frac{5}{8}$
- b. $\frac{29}{80}$
- c. $\frac{1}{16}$ ✓
- d. $\frac{13}{80}$

$$P(X \text{ and } Y) = \frac{1}{8} \times \frac{4}{8} = \frac{4}{64} = \frac{1}{16}.$$

121 Event X has probability $\frac{5}{6}$ and independent event Y has probability $\frac{5}{8}$. Find $P(X \text{ and } Y)$.

- a. $\frac{149}{240}$
- b. $\frac{35}{24}$
- c. $\frac{101}{240}$
- d. $\frac{25}{48}$ ✓

$$P(X \text{ and } Y) = \frac{5}{6} \times \frac{5}{8} = \frac{25}{48}.$$

122 Event X has probability $\frac{1}{7}$ and independent event Y has probability $\frac{4}{5}$. Find $P(X \text{ and } Y)$.

- a. $\frac{1}{70}$
- b. $\frac{4}{35}$ ✓
- c. $\frac{33}{35}$
- d. $\frac{3}{14}$

$$P(X \text{ and } Y) = \frac{1}{7} \times \frac{4}{5} = \frac{4}{35}.$$

123 Event X has probability $\frac{2}{3}$ and independent event Y has probability $\frac{4}{7}$. Find $P(X \text{ and } Y)$.

- a. $\frac{26}{21}$
- b. $\frac{101}{210}$
- c. $\frac{8}{21}$ ✓
- d. $\frac{59}{210}$

$$P(X \text{ and } Y) = \frac{2}{3} \times \frac{4}{7} = \frac{8}{21}.$$

124 Find the mean and median of 16, 30, 32, 39, 38.

- a. 33, 32
- b. 32, 31
- c. 31, 32 ✓
- d. 31, 34

$$\text{Ordered: } 16, 30, 32, 38, 39. \text{ Mean} = \frac{16 + 30 + 32 + 38 + 39}{5} = \frac{155}{5} = 31. \text{ Median (3rd value)} = 32.$$

125 Find the mean and median of 24, 21, 34, 10, 26.

- a. 23, 26
- b. 23, 24 ✓
- c. 25, 24
- d. 24, 23

Ordered: 10, 21, 24, 26, 34. Mean = $\frac{10 + 21 + 24 + 26 + 34}{5} = \frac{115}{5} = 23$. Median (3rd value) = 24.

126 Find the mean and median of 24, 20, 18, 5, 23.

- a. 18, 20 ✓
- b. 20, 20
- c. 20, 18
- d. 18, 22

Ordered: 5, 18, 20, 23, 24. Mean = $\frac{5 + 18 + 20 + 23 + 24}{5} = \frac{90}{5} = 18$. Median (3rd value) = 20.

127 Find the mean and median of 27, 24, 37, 38, 14.

- a. 27, 28
- b. 30, 27
- c. 28, 29
- d. 28, 27 ✓

Ordered: 14, 24, 27, 37, 38. Mean = $\frac{14 + 24 + 27 + 37 + 38}{5} = \frac{140}{5} = 28$. Median (3rd value) = 27.

128 Find the mean and median of 29, 14, 5, 32, 20.

a. 20, 22

b. 23, 20

c. 20, 20 ✓

d. 22, 20

Ordered: 5, 14, 20, 29, 32. Mean = $\frac{5 + 14 + 20 + 29 + 32}{5} = \frac{100}{5} = 20$. Median (3rd value) = 20.